

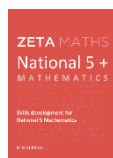
Revision topic list for Form III Mathematics Exam

Use the textbook chapters and any questions suggested by your teacher to practise the skills you have learned in Form III.

Zeta textbook

<https://ebooks.zetapress.co.uk/yyrtww>

Password is Dlrac



Topic	<i>You should be able to...</i>	Chapter	Questions	Revised?
Expanding brackets	Expand two or more single brackets and simplify	5.3	Ex 5.3 Any questions	
	Expand double brackets	5.4 – 5.8	Ex 5.4 to 5.8 Left columns	
	Expand double brackets – binomial and trinomial	5.11	Ex 5.11 Middle column	
Solving equations and inequations	Form linear equations given information	14.7	Ex 14.7 Q18 to Q24	
	Solve equations with fractions	14.9	Third column of Ex 14.9A and 14.9B	
	Solve inequalities	14.8, 14.10	Any from Ex 14.8D and 14.10	
Right-angled trigonometry	Calculate a side in a right-angled triangle given another side and an angle, including calculating the hypotenuse	Zeta N4 Ch21	Any from Ex 21.2	
	Calculate an angle in a right-angled triangle given two sides		Any from Ex 21.3	
	Solve problems involving the use of right-angled trigonometry		All of Ex 21.7A	
Indices I	Use the first three laws of indices to simplify expressions	3.5 – 3.8	Ex 3.7 Q6 to Q16 Ex 3.8 Q11 to 15, 26 to 30	
Standard Form	Carry out calculations with numbers in standard form, <i>with and without a calculator</i>	4.2 4.3	Review Ex 4.4 All questions	
Factorising I	Factorise expressions using the highest common factor	6.1 – 6.3	Ex 6.3 Q10 to 18	
	Identify a difference of two squares and factorise these expressions	6.4	Ex 6.4 Q11 to 20	
	Factorise using both the HCF and a difference of two squares	6.5	Ex 6.5 Q1 to 12	
	Factorise trinomial expressions (unitary coefficient of x^2)	6.6 – 6.9	Ex 6.9 Q1 to 20	
	Identify which technique to use to factorise expressions	6.11	Ex 6.11, Q1,3,8,9,11	
Quadratics I	Solve quadratic equations by factorising	17.4, 17.5A – C	Ex17.5C Q1 to 9	
Percentages	Increase or decrease quantities by percentages	24.4a 24.4b	Ex 24.4B All questions	
	Use the terms appreciation and depreciation and calculate the compound interest over a given number of years			

	Calculate the original amount given the new amount (reverse percentages)	24.5	Ex24.5A, 24.5B Any questions	
--	--	------	---------------------------------	--

Fractions	Add and subtract mixed number fractions	25.3 – 25.5	Review Ex 25.10 All questions	
	Multiply and divide mixed number fractions	25.6 – 25.9		
Statistics	Find the quartiles and calculate the interquartile range of a data set	26.2	Ex 26.2 Q2 (e) to (j)	
Straight Line	Calculate the gradient of the line between two points using the gradient formula	9.2	Ex 9.2 Any from Q1, Q2	
	Use the fact parallel lines have the same gradient to solve problems	9.3A	Ex 9.3A Q1 and Q2 only	
	Sketch a line given its equation	12.2	Ex 12.2 Q1 to 4	
	Find the equation of a straight line given two points, $y = mx + c$ or $y - b = m(x - a)$	12.3	Ex 12.3 Q2 and Q3	
	Find the gradient and y-intercept of a line by rearranging the equation of the lines	12.4	Ex 12.4 Any from Q1	
	Find the equation of a straight line parallel to another	12.5	Ex 12.5 Q1 to 5	
Changing the Subject of Formula	Change the subject of formula involving powers or roots	16.4	Ex 16.4 Q4, 6, 7, 11	
	Change the subject of formula involving multiple operations	16.5 – 16.8	Ex 16.5 Q4, 9, 12 Ex 16.6 Q2, 3, 6, 9 Ex 16.8 Q1, 2, 4	
Rounding	Round numbers to 1, 2 or 3 decimal places	1.1	Ex 1.1 Q1 a to j	
	Round numbers to 1, 2 or 3 significant figures	1.2	Ex 1.2 Q1 a to h	
Circles – arcs and sectors	Calculate the length of an arc of a circle	10.4	Ex 10.4 Any questions	
	Calculate the area of a sector of a circle	10.5	Ex 10.5 Any questions	
	Use algebraic skills to calculate the angle, radius or diameter given the arc length or sector area	10.6	Review Ex 10.7 Q3 to 5	
Properties of Shapes	Calculate the interior or exterior angle in polygons	18.1	Ex18.1A Q1	
	Use the three circle angle properties to find unknown angles in circles	18.2	Ex18.2B Q1 to 6	
Pythagoras' theorem	Use the Converse of Pythagoras' theorem to determine whether a triangle is right-angled	19.2	Ex 19.2 Any questions	
	Use Pythagoras' theorem to solve problem involving circles	19.3	Ex 19.3A Any questions	
	Use Pythagoras' theorem to solve problems involving 3D shapes	19.5	Ex 19.5 Q1	
Volume of 3D shapes	Calculate the volume of prisms and cylinders	11.1 – 11.2A	Ex 11.7 All questions	
	Calculate the volume of part cylinders	11.2B		
	Volume of pyramids, cones, spheres and hemispheres	11.3 – 11.5		
	Calculate the volume of composite 3D shapes	11.7		
	Use algebraic skills to calculate height or radius of a 3D shape given its volume	11.6	Ex 11.6 Any questions	